Inside the Earth | Poster presentation script

Suggested delivery: about 7–8 minutes including pauses to point to the tables and figures. This is a rehearsal estimate, not a course rule. Follow the five columns from left to right.

Opening | one connected evidence chain

Hello. We are the DSC5002 project team at City University of Hong Kong. Inside the Earth asks how earthquake patterns change when we examine geography, time, depth and magnitude together. Please read the five columns from left to right. Each moves from a question, through its evidence, to an interpretation and a limitation. We describe recorded patterns in one fixed catalogue; we do not predict future earthquakes.

Column 1 | scope and choices

Table 1 defines the common sample: 46,463 unique USGS earthquakes from 2000 to 2025, all with preferred magnitude at least five. We retain four negative reported depths and exclude non-earthquake types. This matters: a companion all-type extract contains sixty volcanic, nuclear-explosion or landslide records that do not belong in this earthquake sample. Figure 1 establishes the global setting. Medians, tail shares and rank correlations capture different features. Download completeness does not establish complete detection, and clustered observations are not independent events for significance testing.

Column 2 | concentration and sensitivity

Figure 2 and Table 2 compare boundary distances against equal-area and latitude-controlled reference points. Within one hundred kilometres, the event share is 70.15 percent, versus 9.42 percent for equal-area locations and 10.50 percent for the latitude control. The observed-to-area ratio is 7.45. Figure 3 shows that proximity also varies by region. Table 3 adds a magnitude-threshold check: the nearby share falls to 63.57 percent among events of magnitude seven or more. Broad concentration therefore does not mean identical proximity at every magnitude. These geographic references are not realistic earthquake-occurrence models, and the numerical replicate range is not a confidence interval.

Column 3 | time and measurement comparability

Figure 4 shows fluctuating annual counts, with the peak in 2011. Later-period rates are 6.63 percent lower at magnitude five and 15.42 percent lower at magnitude six, yet the whole-period M5 Sen slope is positive. Table 4 and Figure 5 show why recording composition matters: Mw-family share and horizontal-error availability change substantially. Figure 6 adds the regional contrast: Indonesia and the Philippines decrease, while the Mediterranean and Himalaya group increases. Removing fixed windows around all twenty-eight magnitude-eight-or-larger events reduces the global M5 contrast to minus 3.37 percent. This is a sensitivity check, not declustering or proof of changing physical activity.

Column 4 | regional and local depth structure

Table 5 shows identical thirty-three-kilometre medians for Japan and Kuril and Tonga, Kermadec and Fiji, but deep-event shares of 5.01 and 11.76 percent. Figure 7 pairs route maps with equal-kilometre sections. The local Tonga swath has 186 deep events out of 640, about 29.1 percent, so it cannot represent the complete region. Figure 8 then shows that the depth-distance association varies across regions: about 0.56 in Japan and Kuril but only 0.07 in the Mediterranean and Himalaya group. These selected sections and descriptive associations reveal structure without identifying a causal mechanism.

Column 5 | magnitude, timing and window choice

Table 6 separates event frequency from an illustrative magnitude-based proxy. The twenty-eight events at magnitude eight or more contribute 71.21 percent of the summed proxy, although they are only 0.06 percent of records. Mixed magnitude scales prevent treating this as measured energy or damage. Table 7 compares three cases under identical five-hundred-kilometre and thirty-day windows, excluding anchors. Figure 9 divides by each case's own post-window total: about sixty to seventy-one percent appear in the first three days. Table 8 keeps the radius fixed and varies duration. Counts continue to rise between thirty and ninety days, showing why the window belongs in the result definition. These are nested samples, and time-and-distance membership does not prove causal aftershock status.

Closing | what remains supported

The evidence supports strong descriptive boundary concentration, regional differences in depth, and early concentration within selected sequences. It also shows that magnitude thresholds, reporting composition and event windows affect the comparison. Our contribution is to make this chain of evidence visible and reproducible while keeping its limits clear. Thank you for your attention. We would be happy to discuss a particular sensitivity check.

Discussion prompts

Why compare several thresholds and windows? They define different eligible samples. Reporting their effects makes interpretation depend on explicit choices rather than a single convenient setting.

Why not fit a predictive b-value? An M5 extraction threshold does not establish completeness, and preferred magnitude scales are mixed.

Are the references a statistical null model? No. They control geometric area and latitude, not the tectonic processes governing earthquake occurrence.

Presentation reference

CityU Language Centre: poster script sample. The opening, signposting, methods, findings and closing structure follow the sample; the research content and wording are original.